

2 **(short)** A connection between two pipes of cross sectional areas A_1 and A_2 can be made (i) with a smooth, long transition, where mixing is negligible, or (ii) with a sudden expansion, as indicated in Figs. 2(a) and 2(b), respectively. An incompressible fluid of density ρ flows at velocity V_1 through A_1 in each case. By mass conservation, the velocity through A_2 , far enough from the connection, is $V_2 = (A_1/A_2) V_1$. In case (ii), the pressure on the back-face of the expansion can be approximated as being equal to the upstream pressure p_1 .

- (a) Find the change in pressure between sections 1 and 2 for case (i), and show that it is of the form $\Delta p_a = \frac{1}{2} \rho V_1^2 K_a$, where K_a is only a function of A_1 and A_2 . [5]
- (b) Find the change in pressure between sections 1 and 2 for case (ii), and show that it is of the form $\Delta p_b = \frac{1}{2} \rho V_1^2 K_b$, where K_b is only a function of A_1 and A_2 . [5]

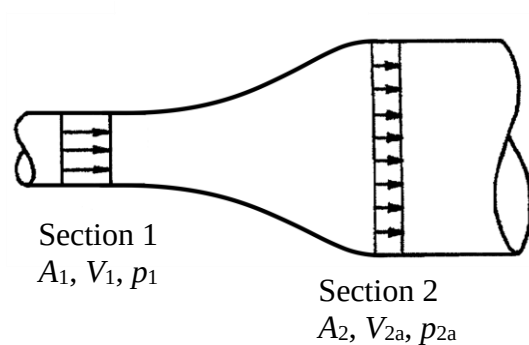


Fig. 2(a)

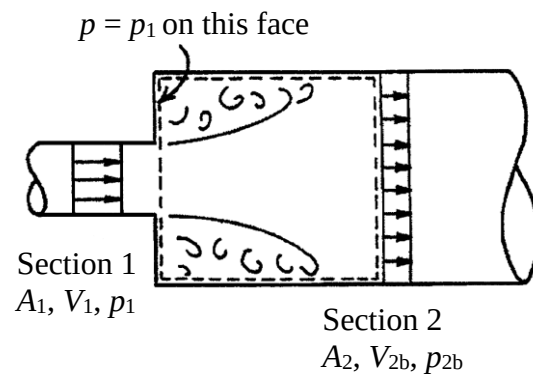


Fig. 2(b)